

CHEBYSHEV–LUCAS CRITICAL POLYNOMIALS: LOCAL BLOCKS, IRREDUCIBILITY AND GALOIS GROUPS

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ABSTRACT. For $d \geq 3$ let $A_d(x) = \sum_{k \geq 0} (-1)^k \binom{d}{2k+1} x^k - 1$, of degree $m = \lfloor (d-1)/2 \rfloor$. Its roots are $\tan^2 \theta$ at the solutions of $\sin d\theta = \sin \theta \cos^{d-1} \theta$, the critical points of the extremal problem of [9], and $A_d + 1$ is the Lucas polynomial $U_d(2, 1+x)$. We conjecture that A_d is irreducible over $\mathbb{Q}$ for all d and develop the tools for this problem. For even d the 2-adic Newton polygon proves irreducibility whenever $\gcd(v_2(d), d/2 - 1) = 1$ (about 94.5% of even d). For odd d we exclude rational roots, prove irreducibility for $d = 2p + 1$ with 2 a primitive root mod p , and show, via an explicit primitive divisor theorem, that factors of any fixed degree disappear for large d . The central new tool is a local block lemma: for $\ell^a \parallel d - 1$ the ℓ -adic Newton polygon has vertices $((\ell^b - 1)/2, a - b)$, the reduction is $x^{(\ell^a - 1)/2} H$, and H is separable iff ℓ does not divide an explicit resultant N_k ; for even d there is a dual block at every odd prime power $q^b \parallel d$, $b \geq 1$, with no such condition. With these we compute Galois groups: $\text{Gal}(A_{2p+1}) \cong S_p$ when 2 is a primitive root mod p ; $\text{Gal}(A_{2p+2}) \supseteq A_p$ whenever $p + 1 \notin \{2^s, 3 \cdot 2^s\}$; and $\text{Gal}(A_{2p+2}) \cong S_p$ for every prime $p \equiv \pm 3 \pmod{8}$, an unconditional infinite family. Computations certify irreducibility for $d \leq 400$ and $\text{Gal}(A_{2p+2}) \cong S_p$ for all primes $p \leq 3000$. All roots of A_d are real, and irreducibility of A_d determines the degree of the sharp constant: $[\mathbb{Q}(C_d) : \mathbb{Q}] = d - 1$ for odd d and $d/2 - 1$ for even d . We explain why local methods alone cannot settle odd prime d .

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1. INTRODUCTION

The polynomials. For an integer $d \geq 3$ put $m = \lfloor (d-1)/2 \rfloor$ and

$$(1) \quad A_d(x) = \sum_{k=0}^m (-1)^k \binom{d}{2k+1} x^k - 1, \quad Q_d(y) = y^m A_d\left(\frac{1-y}{y}\right).$$

Thus $A_3 = 2 - x$, $A_4 = 3 - 4x$, $A_5 = x^2 - 10x + 4$, $A_6 = 6x^2 - 20x + 5$. With $x = \tan^2 \theta$,

$$(2) \quad A_d(\tan^2 \theta) + 1 = \frac{\sin d\theta}{\sin \theta \cos^{d-1} \theta},$$

so the roots of A_d correspond to the solutions of $\sin d\theta = \sin \theta \cos^{d-1} \theta$. These are the critical points of $\theta \mapsto \cos^d \theta - \cos d\theta$, whose maximum is the sharp constant $C_d = \|T_d - x^d\|_{\infty, [-1, 1]}$ for multivariate characteristic-function factorization [9]. There $y = (c_d^*)^2$, with c_d^* the maximizing cosine, is shown to be a root of an integer polynomial of degree m with leading coefficient $2^{d-1} - 1$ [9, Prop. 5.2]; dividing the critical equation $U_{d-1}(c) = c^{d-1}$ by c^{d-1} shows that this polynomial is exactly Q_d . The polynomial A_d appears in [9, Remark 5.3], where its irreducibility for all d is posed as an open problem [9, Sec. 10]. Irreducibility of Q_d fixes the degree of y ; a separate field-degree argument then gives the degree of C_d . Equivalently, $A_d + 1 = U_d(2, 1+x)$, where $U_n(P, Q) = (\alpha^n - \beta^n)/(\alpha - \beta)$, $\alpha + \beta = P$, $\alpha\beta = Q$, is the Lucas polynomial. We call A_d the *Chebyshev–Lucas critical polynomials*.

Conjecture 1.1. A_d is irreducible over $\mathbb{Q}$ for every $d \geq 3$. Moreover $\text{Gal}(A_d/\mathbb{Q}) \cong S_m$.

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ORCID: 0009-0000-4230-4414. Companion to [9, 1].

The conjecture has the flavour of classical open irreducibility problems for binomial-type families (Cauchy–Mirimanoff polynomials, truncated binomial expansions) and of recent work on Chebyshev-like sequences [5]. We found no known theorem that covers A_d .

Main results. Write $a = v_2(d)$ for even d . Our results fall into four groups.

Theorem A (even d). *If d is even, every irreducible factor of A_d over $\mathbb{Q}_2$ has degree divisible by $m/\gcd(a, m)$. In particular A_d is irreducible if $\gcd(a, m) = 1$, and also if $d - 1$ is prime. Among even $d \leq 10^5$ these criteria cover 48015 of 49999 values.*

Theorem B (odd d). *Let $d \geq 5$ be odd. Then A_d has no rational root. If A_d has an irreducible factor of degree k , then $d < \max\{2^{2k+1}, 10^{30}(2k)^9\}$. If $d = 2p + 1$ with p prime and 2 a primitive root modulo p , then A_d is irreducible.*

Theorem C (local blocks). *Let ℓ be an odd prime and $\ell^a \parallel d - 1$. The ℓ -adic Newton polygon of A_d on the roots of positive valuation has vertices $((\ell^b - 1)/2, a - b)$, $b = 0, \dots, a$; the first block is tamely and totally ramified of degree $(\ell - 1)/2$ and the others are wild. The reduction is $A_d \equiv x^{(\ell^a - 1)/2} H \pmod{\ell}$ with explicit H , and H is separable iff $\ell \nmid N_k$, $k = (d - 1)/\ell^a$. For even d and an odd prime power $q^b \parallel d$, $b \geq 1$, there is a dual block on the roots of negative valuation (the reversed polygon has vertices $((q^c - 1)/2, b - c)$), and the remaining reduction is always separable. Inertia at such primes acts as an explicit cycle, or product of cycles, and fixes all other roots.*

Theorem D (Galois groups). *Let p be a prime.*

- (1) *If $p \geq 5$ and 2 is a primitive root modulo p , then $\text{Gal}(A_{2p+1}/\mathbb{Q}) \cong S_p$.*
- (2) *If $p + 1 \notin \{2^s, 3 \cdot 2^s\}$, then $\text{Gal}(A_{2p+2}/\mathbb{Q}) \supseteq A_p$.*
- (3) *If $p \equiv \pm 3 \pmod{8}$, then $\text{Gal}(A_{2p+2}/\mathbb{Q}) \cong S_p$. By Dirichlet's theorem this family is infinite.*

Theorem E (the degree of C_d). *All roots of A_d are real and positive. If A_d is irreducible, then $[\mathbb{Q}(C_d) : \mathbb{Q}] = d - 1$ for odd d and $[\mathbb{Q}(C_d) : \mathbb{Q}] = d/2 - 1$ for even d . This holds unconditionally for every d covered by Theorems A and B and for all $d \leq 400$.*

Since $\deg Q_d = m$, [9, Prop. 5.2] gives only $[\mathbb{Q}(y) : \mathbb{Q}] \leq m$; Theorem E determines the degree of C_d itself for these d and reduces it to Conjecture 1.1 in general.

In Theorem D, part (1) is unconditional for each p ; the infinitude of such p is Artin's conjecture for the base 2, known under GRH [8]. Part (3) needs no hypothesis at all. Theorem D is proved without the classification of finite simple groups (CFSG): besides the Jordan–Marggraff theorem we use only Burnside's theorem on groups of prime degree and Bochert's bound on the minimal degree of doubly transitive groups. The minimal degree theorem of Liebeck and Saxl [11], which relies on CFSG, gives a stronger alternative (Remark 6.1).

Computation. Irreducibility is certified for all $3 \leq d \leq 400$ (Section 9). Moreover $\text{Gal}(A_d) \cong S_m$ for $17 \leq d \leq 82$, $\text{Gal}(A_{2p+1}) \cong S_p$ for all primes $3 \leq p \leq 199$, and $\text{Gal}(A_{2p+2}) \cong S_p$ for all primes $3 \leq p \leq 3000$, the last with a per-prime certificate table.

Limits of local methods. For odd d the question is the *macroscopic* range of factor degrees. Periodicity of short cycles mod p (Proposition 7.1) does not help: on explicit progressions of d every fixed finite set of primes sees all short cycles (Proposition 7.2). For odd prime d all local data at the structured primes are compatible with a factorization; see Example 7.3 for $d = 47$. A proof for all d needs a global argument.

Organization. Section 2 collects identities. Section 3 treats even d and Section 4 odd d . Section 5 proves the local block lemmas and Section 6 the results on Galois groups. Section 7 discusses periodicity and the limits of local methods, Section 9 the computations, and Section 10 open problems.

2. STRUCTURE

Put $a_k(x) = A_k(x) + 1 = \sum_i (-1)^i \binom{k}{2i+1} x^i$ and $b_k(x) = \sum_i (-1)^i \binom{k}{2i} x^i$, so that, with $x = -s^2$,

$$(3) \quad (1+s)^k = b_k(x) + s a_k(x).$$

The leading coefficient of A_d is $(-1)^m$ for odd d and $(-1)^m d$ for even d ; moreover $A_d(0) = d - 1$ and $A_d(-1) = 2^{d-1} - 1$.

Proposition 2.1. (1) $2s(A_d(-s^2) + 1) = (1+s)^d - (1-s)^d$ in $\mathbb{Z}[s]$.
 (2) $A_d(x) + 1 = U_d(2, 1+x) = \sum_j (-1)^j \binom{d-1-j}{j} 2^{d-1-2j} (1+x)^j$.
 (3) $\sum_{j=0}^{d-1} (2u-1)^j - u^{d-1} = u^{d-1} A_d(-(u-1)^2/u^2)$.
 (4) $A_d + 1 = c_d \prod_{e|d, e \geq 3} \Psi_e$, where $\Psi_e \in \mathbb{Z}[x]$ is the minimal polynomial of $\tan^2(\pi/e)$, of degree $\varphi(e)/2$, and $c_d \in \mathbb{Q}^\times$. For odd d all $\tan^2(j\pi/d)$ are algebraic integers; for even d the factor Ψ_e is non-monic exactly when $e = 2q^c$ with q an odd prime.

Proof. (1) is (3) for $k = d$. For (2), in (1) put $\alpha = 1+s$, $\beta = 1-s$, so $\alpha + \beta = 2$ and $\alpha\beta = 1+x$. For (3), substitute $s = (u-1)/u$ in (1): $1-s = 1/u$ and $1+s = (2u-1)/u$. For (4), the roots of $A_d + 1$ are $\tan^2(j\pi/d)$, $1 \leq j \leq m$, by (2); the Galois orbits of these numbers are indexed by $e = d/\gcd(j, d)$. For the last statement write $\tan^2(\pi j/e) = -((\zeta - 1)/(\zeta + 1))^2$ with ζ a primitive e -th root of unity. Then $1 + \zeta$ is a unit unless $-\zeta$ has odd prime power order, i.e. unless $e = 2q^c$. $\square$

Lemma 2.2 (real roots). Put $\Phi(\theta) = A_d(\tan^2 \theta) + 1 = \sin d\theta / (\sin \theta \cos^{d-1} \theta)$ on $(0, \pi/2)$. The equation $\Phi = 1$ has exactly m solutions, all simple: one in $(\pi/(2d), \pi/d)$; one in each half of every positive lobe $(2i\pi/d, (2i+1)\pi/d)$ with $i \geq 1$ and $(2i+1)\pi/d < \pi/2$; and, if $d \equiv 1, 2 \pmod{4}$, one in the first half of the last positive lobe, which reaches $\pi/2$. Hence A_d has m distinct positive roots $x_j = \tan^2 \theta_j$, and $\mathbb{Q}(x_j)$ is totally real.

Proof. At the midpoint of a positive lobe $\sin d\theta = 1 > \sin \theta \cos^{d-1} \theta$, so $\Phi > 1$ there, while $\Phi = 0$ at the ends of the lobe. Hence each half of a full positive lobe contains a solution. On $(0, \pi/d)$ we have $\Phi(0^+) = d$, $\Phi(\pi/(2d)) > 1$ and $\Phi(\pi/d) = 0$. For $d = 4k + 1$ the last lobe is $(2k\pi/d, \pi/2)$, its first half, with $\Phi \rightarrow +\infty$ at $\pi/2$. For $d = 4k + 2$ the last lobe ends at $\pi/2$, where again $\Phi \rightarrow +\infty$. Counting over the four classes of d modulo 4 gives at least $1 + 2(k-1) + 1 = m$ ($d = 4k + 1, 4k + 2$), $1 + 2k = m$ ($d = 4k + 3$) and $1 + 2(k-1) = m$ ($d = 4k$) solutions in the stated intervals. Since $\deg A_d = m$, these are all the roots, and they are simple. $\square$

3. EVEN DEGREE

Theorem 3.1. Let d be even, $a = v_2(d)$ and $m = d/2 - 1$. The 2-adic Newton polygon of A_d is the single segment from $(0, 0)$ to (m, a) , and every interior point lies strictly above it. Hence every irreducible factor of A_d over $\mathbb{Q}_2$ has degree divisible by $m/\gcd(a, m)$, and A_d is irreducible over $\mathbb{Q}$ when $\gcd(a, m) = 1$.

Proof. The constant term $d - 1$ is odd and the leading coefficient is $\pm d$. For $1 \leq k < m$ the coefficient is $\pm \binom{d}{2k+1}$, and for odd j , $\binom{d}{j} = \frac{d}{j} \binom{d-1}{j-1}$ has 2-adic valuation at least $a > ak/m$. Dumas' theorem [6] gives the rest. $\square$

If $a \geq 2$ then m is odd, so $\gcd(a, m) = 1$ whenever a is a power of 2; this covers $d \equiv 2 \pmod{4}$ and $d \equiv 4 \pmod{8}$.

Proposition 3.2. If p is an odd prime, then A_{p+1} is Eisenstein at p .

Proof. $A_{p+1}(0) = p$; for $3 \leq j \leq p-2$, $\binom{p+1}{j} = \binom{p}{j} + \binom{p}{j-1} \equiv 0 \pmod{p}$; the leading coefficient is $\pm(p+1)$. $\square$

Definition 3.3. The exceptional even set is $\mathcal{E} = \{d \text{ even} : \gcd(v_2(d), d/2 - 1) > 1 \text{ and } d - 1 \text{ not prime}\}$.

It begins 56, 248, 296, 320, 344, 352, 392, 512, ... and has 1984 elements up to 10^5 . The 2-adic factorization is in fact explicit.

Proposition 3.4. *Let d be even with $a = v_2(d) \geq 2$, and put $g = \gcd(a, m)$ (odd, since m is odd) and $Q = m/g$. Over $\mathbb{Q}_2$,*

$$A_d = \prod_{\psi} F_{\psi},$$

where ψ runs over the irreducible factors of $y^g - 1$ in $\mathbb{F}_2[y]$ and F_{ψ} is irreducible of degree $Q \deg \psi$. Consequently every proper factor of A_d over $\mathbb{Q}$ has degree Qs , where s is the sum of $\deg \psi$ over a proper nonempty set of these ψ . For $g = 3$ the only possible degrees are Q and $2Q$; for $g = 5$ they are Q and $4Q$.

Proof. Apply Ore's theorem on the residual polynomial [14, 7] to $x^m A_d(1/x)$, whose Newton polygon is the single segment from $(0, a)$ to $(m, 0)$ (Theorem 3.1). Its lattice points are at the abscissae iQ , $0 \leq i \leq g$, and all interior coefficients lie strictly above the segment, so the residual polynomial is $c_0 + c_g y^g$ with $c_0 \equiv d/2^a \equiv 1$ and $c_g \equiv d - 1 \equiv 1 \pmod{2}$, i.e. $y^g + 1$. It is separable because g is odd, so A_d is regular and its $\mathbb{Q}_2$ -factors correspond to the irreducible factors of $y^g + 1 = y^g - 1$ over $\mathbb{F}_2$, each of degree Q times the degree of the factor. $\square$

Remark 3.5. For $d \in \mathcal{E}$, the degrees allowed by Proposition 3.4 can be tested against the local data at the other structured primes: the dual blocks and the separable reduction at each odd $q \mid d$ (Lemma 5.4), and the blocks and H at each odd $\ell \mid d - 1$ (Lemma 5.1). For the 15 elements of $\mathcal{E}$ below 1000 this excludes every proper degree except for $d = 248$ (degrees 41, 82) and $d = 352$ (degrees 35, 140), which are settled by the general certificates of Section 9. The exclusions depend on the factorization patterns of these reductions, so they give no uniform argument for $\mathcal{E}$.

For $d = \ell^c + 1$ with $c \geq 2$, Lemma 5.1 shows that the ℓ -adic polygon is never a single segment, so Proposition 3.2 does not extend to prime powers.

4. ODD DEGREE

Lemma 4.1. *For odd d , $A_d(x - 1) \equiv x^m - 1 \pmod{2}$.*

Proof. Modulo 2 we have $(2u - 1)^j \equiv 1$, so Proposition 2.1(3) gives $u^{2m} \bar{A}(-(u - 1)^2/u^2) = 1 + u^{2m}$ in $\mathbb{F}_2[u]$, where $\bar{A} = A_d \pmod{2}$. Since $(u + 1)^{2k} = (u^2 + 1)^k$ in characteristic 2, with $U = u^2$ this reads $U^m \bar{A}(1 + 1/U) = 1 + U^m$, i.e. $\bar{A}(1 + Y) = 1 + Y^m$. $\square$

Lemma 4.2. *Let d be odd, x a root of A_d , $s^2 = -x$, $\alpha = 1 + s$, $\beta = 1 - s$, $\gamma = \alpha/\beta$, $K = \mathbb{Q}(s)$. Then α, β are units at every prime of K above 2, and γ is not a root of unity.*

Proof. All these numbers are algebraic integers, because A_d is monic up to sign. Let $\mathfrak{P} \mid 2$. By Lemma 4.1, $A_d(t - 1) = t^m - 1 + 2R(t)$ with $R \in \mathbb{Z}[t]$, so $t = 1 + x = \alpha\beta$ satisfies $t^m \equiv 1 \pmod{\mathfrak{P}}$ and α, β are $\mathfrak{P}$ -units. Hence $\gamma + 1 = 2/\beta$ has $\nu_{\mathfrak{P}}(\gamma + 1) = \nu_{\mathfrak{P}}(2) > 0$. Suppose γ is a root of unity. Then $\gamma \neq -1$, since $\alpha + \beta = 2$, so $\eta = -\gamma \neq 1$ is a root of unity with $\nu(1 - \eta) = \nu(2)$. If the order of η is not a power of 2, then $1 - \eta$ is a 2-unit. If the order is 2^c , then $\nu(1 - \eta) = \nu(2)/2^{c-1}$, which equals $\nu(2)$ only for $c = 1$, i.e. $\gamma = 1$ and $x = 0$. But $A_d(0) = d - 1 \neq 0$. $\square$

Proposition 4.3. *For odd $d \geq 5$, A_d has no rational root.*

Proof. A rational root is an integer $r \mid d - 1$. For $z \geq 0$, $A_d(-z) = \sum_k \binom{d}{2k+1} z^k - 1 \geq d - 1 > 0$, so $r > 0$. By Lemma 4.1, $A_d(r) \equiv (r + 1)^m + 1 \pmod{2}$, so r is even. Then $(1 \pm \sqrt{-r})$ is a Lucas pair: $\alpha + \beta = 2$ and $\alpha\beta = 1 + r$ are coprime, and α/β is not a root of unity by Lemma 4.2. By Proposition 2.1(1), $A_d(r) = 0$ iff $u_d(\alpha, \beta) = 1$. For $d > 30$ the number u_d has a primitive divisor [3, Thm 1.4], so $|u_d| > 1$. For $5 \leq d \leq 30$ one checks the divisors of $d - 1$ directly. $\square$

Theorem 4.4. *Let p be an odd prime with $\text{ord}_p(2) = p - 1$ and $d = 2p + 1$. Then A_d is irreducible.*

Proof. Here $m = p$ and $A_d(x - 1) \equiv (x - 1)\Phi_p(x) \pmod{2}$, where Φ_p is irreducible over $\mathbb{F}_2$ because 2 has order $p - 1$. The leading coefficient is odd, so a proper factorization over $\mathbb{Z}$ would have a factor of degree 1, i.e. a rational root. This contradicts Proposition 4.3. $\square$

Theorem 4.5 (small factors). *Let d be odd. If A_d has an irreducible factor of degree k over $\mathbb{Q}$, then $d < \max\{2^{2k+1}, 10^{30}(2k)^9\}$.*

Proof. Let x be a root of such a factor, with s, α, β, γ as in Lemma 4.2. Since $s = (\gamma - 1)/(\gamma + 1)$, $\mathbb{Q}(\gamma) = \mathbb{Q}(s)$ has degree $D \leq 2k$, and $\beta = 2/(\gamma + 1)$ is an algebraic integer of $\mathbb{Q}(\gamma)$. Also $x \neq 0, -1$, so $\gamma \neq 0, 1$. By Proposition 2.1(1), $\alpha^d - \beta^d = \alpha - \beta$, i.e. $\gamma^d - 1 = (\gamma - 1)\beta^{1-d}$. Hence for every prime $\mathfrak{p}$ of $\mathbb{Q}(\gamma)$,

$$\nu_{\mathfrak{p}}(\gamma^d - 1) = \nu_{\mathfrak{p}}(\gamma - 1) + (1 - d)\nu_{\mathfrak{p}}(\beta) \leq \nu_{\mathfrak{p}}(\gamma - 1),$$

so $\gamma^d - 1$ has no primitive divisor. Since γ is not a root of unity (Lemma 4.2), the explicit form [4, Thm 4.2] of Schinzel's theorem [15] gives $d < \max\{2^{D+1}, 10^{30}D^9\}$. $\square$

Remark 4.6. The argument uses that the constant is -1 . For $U_d(2, t) - c$ one gets $\gamma^d - 1 = c(\gamma - 1)\beta^{1-d}$, and a primitive divisor may divide c . Numerically, $U_d(2, t) - c$ is irreducible for $c \in \{\pm 1, \pm 2, 3, 4\}$ and $d \leq 50$.

5. LOCAL BLOCKS

Put $N_k = \prod_{\zeta^k=1, \zeta \neq 1} ((2 - \zeta)^k - 1) \in \mathbb{Z} \setminus \{0\}$; it is nonzero because $|2 - \zeta| > 1$ for $\zeta \neq 1$ on the unit circle. Equivalently, $N_k = \text{Res}((x^k - 1)/(x - 1), (2 - x)^k - 1)$, and $\text{Res}(R_k, S_k) = \pm N_k/k$ for $R_k(z) = ((1 + z)^k - 1)/z$ and $S_k(z) = ((1 - z)^k - 1)/z$.

Lemma 5.1 (block at $\ell^a \parallel d - 1$). *Let ℓ be an odd prime, $d = k\ell^a + 1$ with $\ell \nmid k$, $N = \ell^a$, $E = (N - 1)/2$ and $E_b = (\ell^b - 1)/2$.*

- (A) $v_{\ell}\binom{d}{2i+1} = a - v_{\ell}(i(2i + 1))$ for $1 \leq i \leq E$.
- (B) *The ℓ -adic Newton polygon of A_d on its positive-valuation part has vertices $(E_b, a - b)$, $b = 0, \dots, a$. Its segments have height 1 and lengths $\ell^{b-1}(\ell - 1)/2$. The first block is tamely and totally ramified of degree $e = (\ell - 1)/2$; for $b \geq 2$ the blocks are wild.*
- (C) $A_d \equiv x^E H \pmod{\ell}$, with $H = (-1)^E a_k^N + x^{E+1} w^N$, $w = (b_k - 1)/x$ and $H(0) \equiv (-1)^E k \not\equiv 0$.
- (D) H is separable iff $\ell \nmid N_k$, iff there are no $\zeta_1 \neq \zeta_2$ in $\mu_k(\overline{\mathbb{F}}_{\ell})$ with $\zeta_1 + \zeta_2 = 2$.

Proof. (A) We have $\binom{d}{j} = \frac{d}{j} \binom{d-1}{j-1}$ and $v_{\ell}(d) = 0$. For $1 \leq t < N$, $v_{\ell}\binom{kN}{t} = a - v_{\ell}(t)$, since $\binom{kN-1}{t-1} \not\equiv 0$ by Lucas' theorem. Finally $j = 2i + 1$ and $j - 1 = 2i$ are not both divisible by ℓ . (B) A point with $v_{\ell}(i(2i + 1)) = t$ has $i \geq E_t$, so it lies on or to the right of the vertex $(E_t, a - t)$ at the same height. The slopes $-2/(\ell^{b-1}(\ell - 1))$ increase. Each segment has height 1, so each block is irreducible over $\mathbb{Q}_{\ell}^{\text{ur}}$ and totally ramified, of degree $\ell^{b-1}e$. (C) In $\mathbb{F}_{\ell}[s]$, $(1 \pm s)^d = (1 \pm s)(1 \pm s^N)^k$. By (3) and $a_k(x^N) = a_k(x)^N$, $A_d + 1 \equiv (-x)^E a_k^N + b_k^N$, and $b_k - 1 = xw$. (D) Modulo ℓ , $H' \equiv (E + 1)x^E w^N$ with $E + 1 = (N + 1)/2 \not\equiv 0$. A repeated root r of H has $r \neq 0$, hence $w(r) = 0$ and then $a_k(r) = 0$. With $s^2 = -r$ this means $(1 \pm s)^k = 1$; conversely such s gives a repeated root. Reducing μ_k modulo a prime above ℓ gives the resultant form. $\square$

Lemma 5.2 (tame subfield). *In Lemma 5.1, let $K_0 = \mathbb{Q}_{\ell}^{\text{ur}}$ and $T = K_0(\ell^{1/e})$. Then $K_0(\beta) \supseteq T$ for every root β of every block.*

Proof. $K_0(\beta)/K_0$ is totally ramified of degree $\ell^{b-1}e$. Its maximal tamely ramified subextension has degree equal to the prime-to- ℓ part of the ramification index, namely e [17, Ch. IV]. Since the residue field of K_0 is algebraically closed and $\ell \nmid e$, T is the only tamely ramified extension of K_0 of degree e . $\square$

Corollary 5.3 (inertia at ℓ). *In Lemma 5.1:*

- (1) *if $a = 1$ and $\ell \nmid N_k$, inertia at ℓ contains an e -cycle that fixes every other root;*
- (2) *if $a \geq 2$ and $\ell \nmid N_k$, an element of order e that generates the tame part of inertia is a product of $(\ell^a - 1)/(\ell - 1)$ disjoint e -cycles on the block roots and fixes the other roots; no inertia element is a single e -cycle;*
- (3) *inertia at ℓ contains an odd permutation iff $\ell \equiv 1 \pmod{4}$ and a is odd.*

Proof. If $\ell \nmid N_k$, then by Hensel the roots outside the blocks lie in K_0 and are fixed by inertia. Now T/K_0 is cyclic of degree e and acts simply transitively on the roots of the first block. By Lemma 5.2, an element that is nontrivial on T fixes no root of any block, which gives (1) and (2). For (3), wild inertia has odd order ℓ^r , and the number of e -cycles is $\equiv a \pmod{2}$. $\square$

Lemma 5.4 (dual block). *Let d be even, q an odd prime, $q^b \parallel d$, $N = q^b$, $E = (N - 1)/2$ and $e = d/N$.*

- (1) $A_d \equiv (-x)^E a_e(x)^N - 1 \pmod{q}$, a polynomial of degree $m - E$.
- (2) *This reduction is separable.*
- (3) *The q -adic polygon of $x^m A_d(1/x)$ has vertices $((q^c - 1)/2, b - c)$, $c = 0, \dots, b$, on its positive-slope part. Thus A_d has exactly E roots of negative valuation, forming blocks of degrees $q^{c-1}(q - 1)/2$.*

Consequently inertia at q fixes all roots of non-negative valuation; for $b = 1$ and $q \geq 5$ it contains an E -cycle fixing all other roots; for $b \geq 2$ its elements move at most E roots; and it contains an odd permutation iff $q \equiv 1 \pmod{4}$ and b is odd.

Proof. (1) In $\mathbb{F}_q[s]$, $(1 \pm s)^d = (1 \pm s^N)^e = b_e(x)^N \pm s^N a_e(x)^N$. Take the difference and divide by $2s$. (2) The derivative is $\equiv -E(-x)^{E-1} a_e^N$, which is nonzero at every root, since a root has $x \neq 0$ and $a_e(x) \neq 0$. (3) The coefficient of x^j in the reversed polynomial is $\pm \binom{d}{2j+1}$, and $v_q\left(\binom{eN}{t}\right) = b - v_q(t)$ for $1 \leq t < N$ while $v_q\left(\binom{eN}{N}\right) = 0$; argue as in Lemma 5.1(B). The inertia statements follow from Lemma 5.2 and Corollary 5.3 applied to $y = 1/x$. $\square$

For odd d the leading coefficient is ± 1 ; the reduction (1) then has full degree and is separable, so $q \mid d$ is unramified.

Lemma 5.5 (support bound). *Let ℓ be prime. Take $f = A_d$ if ℓ does not divide its leading coefficient and $f = x^m A_d(1/x)$ otherwise. Write $\bar{f} = c \prod_i \phi_i^{e_i}$ over $\mathbb{F}_\ell$ and put $M_\ell(d) = \sum_{e_i \geq 2} e_i \deg \phi_i$. Then every element of every inertia group above ℓ moves at most $M_\ell(d)$ roots of A_d .*

Proof. A simple root of $\bar{f}$ lifts by Hensel to exactly one root of f in an unramified extension; these $m - M_\ell$ roots are fixed by inertia. $\square$

6. GALOIS GROUPS

We use the following facts about a transitive group $G \leq S_n$.

- (a) If n is prime, G is primitive.
- (b) (Jordan–Marggraff) If G is primitive and contains a subgroup with an orbit of length k , $1 < k < n/2$, fixing the remaining points, then $G \supseteq A_n$ [19, Thms 13.4–13.5]; see [12] for a short proof of Marggraff’s theorem and [10] for its history.
- (c) (Jones) If G is primitive and contains a cycle fixing at least three points, then $G \supseteq A_n$ [10, Cor. 1.3].
- (d) (Burnside–Bochert) If $n = p \geq 41$ is prime and G contains $g \neq 1$ moving fewer than $p/4$ points, then $G \supseteq A_p$.

For (d): by Burnside’s theorem a transitive group of prime degree p that is not doubly transitive is a subgroup of $\text{AGL}(1, p)$ [13]. A nonidentity element of $\text{AGL}(1, p)$ moves at least $p - 1$ points, so g excludes this case. By Schomburg’s classification-free form of Bochert’s theorem [16, Thm 3.1] (see also [2]), a doubly transitive group of degree $n \geq 38$ not containing A_n has minimal degree at least $n/4$. Items (a), (b), (d) do not use CFSG; (c) does, and it is not needed in the proofs below.

Remark 6.1 (Liebeck–Saxl). Using CFSG, a primitive group of minimal degree $< n/3$ has $n = \binom{m'}{k}^r$ and $A_{m'}^r \leq G \leq S_{m'} \wr S_r$ in the product action on k -sets [11]. For $n = p$ prime this forces $r = 1$ and $k \in \{1, m' - 1\}$. Hence in (d) one may replace $p/4$ by $p/3$ and drop $p \geq 41$. This is needed only in one subcase of Theorem 6.4(2), which Theorem D does not use.

Theorem 6.2. *Let $p \geq 5$ be prime, $d = 2p + 1$, and suppose A_d is irreducible. Then $\text{Gal}(A_d) \supseteq A_p$. If moreover $p \equiv 1 \pmod{4}$ or $p \equiv 3 \pmod{8}$, then $\text{Gal}(A_d) \cong S_p$.*

Proof. G is transitive of prime degree p , hence primitive. Apply Lemma 5.1 with $\ell = p$, $a = 1$, $k = 2$: here $N_2 = 8$, so the block is clean for every odd p . (Directly, $A_d \equiv (-x)^e((-x)^{e+1} + 2) \pmod{p}$ with $e = (p - 1)/2$.) Corollary 5.3(1) gives an e -cycle, $1 < e < p/2$, fixing $e + 1 \geq 3$ roots,

and (b) gives $A_p \subseteq G$. If $p \equiv 1 \pmod{4}$, the e -cycle is odd. If $p \equiv 3 \pmod{8}$, then $(2/p) = -1$, so $f = \text{ord}_p(2)$ is even and $(p-1)/f$ is odd. The Frobenius at 2 (Lemma 4.1, $m = p$ odd) has one fixed point and $(p-1)/f$ cycles of length f , so it is odd. $\square$

Corollary 6.3. *If $p \geq 5$ is prime and $\text{ord}_p(2) = p-1$, then $\text{Gal}(A_{2p+1}/\mathbb{Q}) \cong S_p$.*

Proof. Irreducibility is Theorem 4.4. The Frobenius at 2 has cycle type $(1, p-1)$, which is odd. $\square$

Theorem 6.4. *Let $p \geq 5$ be prime and $d = 2p+2$ (so $m = p$); A_d is irreducible by Theorem 3.1.*

- (1) *If $p+1 \notin \{2^s, 3 \cdot 2^s\}$, then $\text{Gal}(A_d) \supseteq A_p$.*
- (2) *If some $\ell^a \parallel 2p+1$ has $\ell^a \geq 5$, $k = (2p+1)/\ell^a \geq 3$ and $\ell \nmid N_k$, then $\text{Gal}(A_d) \supseteq A_p$.*

Proof. (1) Suppose $p+1$ has a prime $q \geq 5$ to the first power. Then $q \parallel d$, and Lemma 5.4 gives an E -cycle, $2 \leq E = (q-1)/2 < p/2$, fixing at least three roots, so (b) applies. Otherwise some odd $q^b \parallel p+1$ has $b \geq 2$. Its inertia is nontrivial, since the polygon has a segment of non-integral slope, and moves at most $E = (q^b-1)/2 \leq (p-1)/4 < p/4$ roots. For $p \geq 41$, (d) applies. The only prime $p < 41$ in this case is $p = 17$; there $2p+1 = 35$, and part (2) applies with $\ell = 7$, $a = 1$, $k = 5$, $N_5 \not\equiv 0 \pmod{7}$ (Jordan–Marggraff). (2) If $a = 1$, Corollary 5.3(1) gives an e -cycle with $2 \leq e < p/2$ fixing at least three roots. If $a \geq 2$, inertia is nontrivial and moves at most $E = (\ell^a-1)/2$ roots by Lemma 5.5. Now $E < p/4$ iff $k \geq 5$, so (d) applies when $k \geq 5$ and $p \geq 41$. For $k = 3$ we have $E = (p-1)/3 < p/3$, and Remark 6.1 applies (this subcase uses CFSG). $\square$

Theorem 6.5 (2-adic square class). *Let d be even with $a = v_2(d) \geq 2$ and $\gcd(a, m) = 1$. Then $\text{disc}(A_d) \in (\mathbb{Q}_2^\times)^2$ iff $m \equiv \pm 1 \pmod{8}$.*

Proof. By Theorem 3.1 all roots have valuation $-a/m$, so $K = \mathbb{Q}_2(\text{root})$ is totally and tamely ramified of odd degree m . Such a field is $\mathbb{Q}_2(\varpi^{1/m})$ for a uniformizer $\varpi = 2u$ [17, Ch. IV]. Every unit of $\mathbb{Z}_2$ is an m -th power: this is clear for ± 1 , and on $1+4\mathbb{Z}_2 \cong \mathbb{Z}_2$ the map $v \mapsto mv$ is bijective. Hence $K = \mathbb{Q}_2(2^{1/m})$. The reversed polynomial has unit leading coefficient $d-1$ and the same discriminant, so modulo squares $\text{disc}(A_d) \equiv \text{disc}(\mathcal{O}_K) \equiv \text{disc}(x^m - 2) = (-1)^{(m-1)/2} m^m 2^{m-1} \equiv (-1)^{(m-1)/2} m$. This is a 2-adic square iff it is $\equiv 1 \pmod{8}$, i.e. iff $m \equiv \pm 1 \pmod{8}$. $\square$

Corollary 6.6. *For every prime $p \equiv \pm 3 \pmod{8}$, $\text{Gal}(A_{2p+2}/\mathbb{Q}) \cong S_p$.*

Proof. Here $v_2(2p+2) \geq 2$. If $p \equiv 5 \pmod{8}$ and $p \geq 13$, then $p+1 = 2 \cdot o$ with $o \geq 7$ odd. If $p \equiv 3 \pmod{8}$ and $p \geq 19$, then $p+1 = 4 \cdot o$ with $o \geq 5$ odd. In both cases $p+1 \notin \{2^s, 3 \cdot 2^s\}$, and Theorem 6.4(1) gives $A_p \subseteq G$. Theorem 6.5 with $m = p \equiv \pm 3 \pmod{8}$ gives a non-square discriminant, so $G = S_p$. The remaining primes $p = 3, 5, 11$ are covered by Section 9. $\square$

For the other residues the parity comes from closed-form local square classes. The decomposition group at q lies in A_m iff $\text{disc} \in (\mathbb{Q}_q^\times)^2$. Inertia alone does not detect this: for $d = 64$ and $d = 76$ the odd valuations $v_3 = 7$ and $v_5 = 17$ come from wild blocks. Let χ_q be the Legendre symbol.

Proposition 6.7 (dual square class). *In Lemma 5.4 let $b = 1$, and put $n = m - E$, $\delta = e/2 - 1$, $L = (-1)^\delta e$ and $\bar{D} = -(-1)^{n(n-1)/2 + (n+1)\delta} (-E)^n L^{n-1}$. Then $\text{disc}(A_d) \notin (\mathbb{Q}_q^\times)^2$ iff E is even, or E is odd and $\chi_q((-1)^{(E-1)/2} E) \chi_q(\bar{D}) = -1$.*

Proof. The reversed polynomial R has unit leading coefficient and $\text{disc } R = \text{disc } A_d$. Over $\mathbb{Z}_q$, $R = (d-1)R_B R_C$, where R_B is Eisenstein of degree E and R_C is monic with $\bar{R}_C$ proportional to the reversal of $\bar{A} = (-x)^E a_e^q - 1$. Modulo squares, $\text{disc } R \equiv \text{disc } R_B \text{disc } R_C$.

For the block, $\text{disc } R_B = (-1)^{E(E-1)/2} N(R'_B(\beta))$ and $R'_B(\beta) = E\beta^{E-1}(1+\varepsilon)$ with $v(\varepsilon) > 0$. The norm of $1+\varepsilon$ is a square, so the class is $(-1)^{E(E-1)/2} E^E N(\beta)^{E-1}$. Since $N(\beta)$ has valuation 1, this has odd valuation for even E and class $(-1)^{(E-1)/2} E$ for odd E .

For the rest, $\text{disc } R_C$ is a unit congruent, modulo squares, to $\text{disc } \bar{A}$. In $\mathbb{F}_q[x]$ we have $\bar{A}' = -E(-x)^{E-1} a_e^q$, of degree $n-1$, and $\bar{A} = -1$ at the roots of a_e . So $\text{Res}(\bar{A}, \bar{A}') = (-E)^n (-1)^{E-1} (-1)^{(n+1)\delta} L^n$ and $\text{lc } \bar{A} = (-1)^E L$, which gives $\text{disc } \bar{A} = \bar{D}$. $\square$

Proposition 6.8 (square class at $\ell \parallel d-1$). *Let d be even, $\ell \parallel d-1$, $k = (d-1)/\ell \geq 3$ and $\ell \nmid N_k$. Put $e = (\ell-1)/2$, $n = \ell(k-1)/2$, $\delta_a = (k-1)/2$, $\delta_w = (k-3)/2$, $\lambda = (-1)^{\delta_a}k$, $\lambda_H = (-1)^{e+\delta_a}$ and $\rho = \text{Res}(w, a_k) \bmod \ell \neq 0$. Let*

$$\bar{D}_H = (-1)^{n(n-1)/2} \lambda_H^{e-1} (e+1)^n ((-1)^{n+e}k)^e (-1)^{(n+e)\delta_w} \lambda^{n-\delta_a} \rho.$$

Then $\text{disc}(A_d) \notin (\mathbb{Q}_\ell^\times)^2$ iff e is even, or e is odd and $\chi_\ell((-1)^{(e-1)/2}e)\chi_\ell(\bar{D}_H) = -1$.

Proof. The argument is the same as for Proposition 6.7, with the Eisenstein block of degree e at 0 and H from Lemma 5.1(C). Here $H' \equiv (e+1)x^e w^\ell$ has degree $n-1-e$, so $\text{disc } H = (-1)^{n(n-1)/2} \text{lc}(H)^{e-1} \text{Res}(H, H')$. Moreover

$$\text{Res}(H, x) = (-1)^n H(0), \quad \text{Res}(H, w) = (-1)^{n\delta_w} \lambda^n \prod_{w(r)=0} H(r),$$

with $H(r) = (-1)^e a_k(r)^\ell$ at the roots of w and $\prod_{w(r)=0} a_k(r) = \rho \lambda^{-\delta_a}$. $\square$

Theorem 6.5 and Propositions 6.7, 6.8 agree with the square class of the exact discriminant in 63, 144 and 125 of 63, 144 and 125 cases respectively (all admissible even $d \leq 290$ in the tested ranges).

7. PERIODICITY AND THE LIMITS OF LOCAL METHODS

Proposition 7.1 (periodicity). *Let p be an odd prime and $\ell \geq 1$. The set of nonzero roots of $A_d \bmod p$ in $\mathbb{F}_{p^\ell}^\times$ depends only on $d \bmod (p^{2\ell} - 1)$, and $x = 0$ is a root iff $p \mid d-1$. Hence the number of distinct monic irreducible factors of degree ℓ , other than x , is periodic in d with period dividing $p^{2\ell} - 1$.*

Proof. For $x \in \mathbb{F}_{p^\ell}^\times$ choose $s \in \mathbb{F}_{p^{2\ell}}$ with $s^2 = -x$. By Proposition 2.1(1), $A_d(x) = 0$ iff $(1+s)^d - (1-s)^d = 2s$, which depends only on $d \bmod (p^{2\ell} - 1)$. For $s = \pm 1$ the condition is $2^{d-1} = 1$, which depends on d modulo a divisor of $p-1$. $\square$

Proposition 7.2 (obstruction). *If $d \equiv 1 \pmod{p^{2\ell}-1}$, then every $x \in \mathbb{F}_{p^\ell}^\times$ is a root of $A_d \bmod p$. Hence, for any finite set P of odd primes, on the progression $d \equiv 1 \pmod{\text{lcm}_{p \in P} p(p^{2\ell} - 1)}$ the factorization pattern at every $p \in P$ contains all cycles of length dividing ℓ ; for odd d the reduction mod 2 also has a root. Cycle-type data at finitely many primes can therefore never exclude linear factors for all d , and small factors need global input such as Proposition 4.3 and Theorem 4.5.*

Proof. For $s \in \mathbb{F}_{p^{2\ell}}$ we have $(1 \pm s)^d = 1 \pm s$. $\square$

Example 7.3 ($d = 47$). Here $m = 23$, and the structured primes are 2 and the prime divisors of $d(d-1) = 47 \cdot 46$. Modulo 2 the Frobenius has cycle type $(1, 11, 11)$, since 2 has order 11 modulo 23. At 23 there is a tame Eisenstein block of degree 11 and an unramified rest of degree 12. At $47 = d$, $A_d \equiv (-x)^{23} - 1$ splits completely, so the Frobenius is trivial. All these data are compatible with a factorization of degrees $11 + 12$ in which the degree-11 factor is the 23-adic block. Hence no argument that uses only these primes can prove irreducibility here (it holds by computation). The same holds for $d = 83, 383, 479$. For the composite $d = 2p + 1$ with $\text{ord}_p(2) = (p-1)/2$ and $d \leq 400$, the Frobenius at the primes $q \mid d$ closes every case except $d = 15$.

Remark 7.4 (orbit norms). Let d be odd and $A_d = FG$ in $\mathbb{Z}[x]$. Proposition 2.1(4) gives $\prod_{x^0 \in O_e} F(x^0) = \pm 1$ for every orbit O_e of $\tan^2(j\pi/d)$, because $A_d(x^0) = -1$. This is the algebraic-integer analogue of Schur's argument for $\prod (x - a_i) - 1$. It is, however, automatic (it equals $\text{Res}(\Psi_e, F) \text{Res}(\Psi_e, G) = \pm 1$), so it is only a necessary condition on the root set of a putative factor. For prime d it is a single real equation.

8. THE DEGREE OF THE SHARP CONSTANT

Let $g(\theta) = \cos^d \theta - \cos d\theta$. By [9, Thm 2.4], $C_d = \max_{[0, \pi/d]} g$ is attained at a unique interior critical point; since $g' = d(\sin d\theta - \sin \theta \cos^{d-1} \theta)$, the critical points in $(0, \pi/2)$ are the θ_j of Lemma 2.2, and $C_d = g(\theta_1)$. Put $c_j = \cos \theta_j$ and $y_j = c_j^2$; the y_j are the roots of Q_d .

Lemma 8.1. (1) $g(\theta_j) = U_{d-2}(c_j)$.

(2) If $\cos d\theta_j < 0$ then $1 < g(\theta_j) < 2$; if $\cos d\theta_j > 0$ then $-1 < g(\theta_j) < 0$.

(3) Along the roots with $\cos d\theta_j < 0$ the values $g(\theta_j)$ strictly decrease; along those with $\cos d\theta_j > 0$ the values $|g(\theta_j)|$ strictly increase.

In particular the numbers $|g(\theta_j)|$, $1 \leq j \leq m$, are nonzero and pairwise distinct.

Proof. (1) $\sin d\theta = \sin \theta U_{d-1}(\cos \theta)$, so at a critical point $U_{d-1}(c) = c^{d-1}$; with $T_d = cU_{d-1} - U_{d-2}$ we get $g = c^d - T_d(c) = U_{d-2}(c)$. (2) Put $u = \cos^d \theta$, $t = \tan \theta$ and $s = \sin \theta \cos^{d-1} \theta = ut = \sin d\theta$, so $\cos d\theta = \pm \sqrt{1 - s^2}$. With $x = t^2$ we have $ut^2 = x(1+x)^{-d/2} \leq M_d := \frac{2}{d-2}(1 - \frac{2}{d})^{d/2} \leq \frac{2}{e^{d-2}} < 1$. If $\cos d\theta < 0$ then $g - 1 = u - s^2/(1 + \sqrt{1 - s^2}) \geq u(1 - M_d) > 0$. If $\cos d\theta > 0$ then $g < 0$ because $u^2 + s^2 = \cos^{2d-2} \theta < 1$. (3) By Lemma 2.2, consecutive roots of the same type satisfy $\theta' - \theta > h := 3\pi/(2d)$. Since $\cos \theta' \leq \cos \theta \cos(\theta' - \theta)$, the value $u' = \cos^d \theta'$ satisfies $u'/u \leq \cos^d h \leq e^{-dh^2/2} = e^{-9\pi^2/(8d)}$. For the first type, $g - 1 \in [u(1 - M_d), u]$, and for the second, $1 - |g| \in [u, u(1 + M_d)]$. So both claims follow from

$$e^{-9\pi^2/(8d)} < 1 - \frac{2}{e^{d-2}} \quad (d \geq 3),$$

which also gives $e^{-9\pi^2/(8d)}(1 + M_d) < 1$. For $d \geq 20$ it follows from $e^{-a} \leq 1 - a + a^2/2$ with $a = 9\pi^2/(8d)$; for $3 \leq d \leq 19$ it is checked directly (the margin at $d = 3$ is 0.239). $\square$

Theorem 8.2. If A_d is irreducible, then $[\mathbb{Q}(C_d) : \mathbb{Q}] = d - 1$ for odd d and $[\mathbb{Q}(C_d) : \mathbb{Q}] = d/2 - 1$ for even d .

Proof. Q_d is irreducible of degree m , since A_d is. For even d , U_{d-2} is even, so $C_d = S(y_1)$ with $S \in \mathbb{Q}[y]$. The embeddings of $\mathbb{Q}(y_1)$ send C_d to $S(y_j) = g(\theta_j)$, which are distinct by Lemma 8.1; hence the degree is $m = d/2 - 1$.

For odd d , U_{d-2} is odd. The norm of y_1 is $\prod_j y_j = |Q_d(0)|/(2^{d-1} - 1) = 1/(4^m - 1)$. This is not the square of a rational number, because $4^m - 1$ lies strictly between $(2^m - 1)^2$ and $(2^m)^2$. So $c_1 \notin \mathbb{Q}(y_1)$, $[\mathbb{Q}(c_1) : \mathbb{Q}] = 2m$, and the embeddings send c_1 to all $\pm c_j$. Then $C_d = U_{d-2}(c_1)$ has the $2m$ conjugates $\pm g(\theta_j)$, which are distinct by Lemma 8.1. Hence the degree is $2m = d - 1$. $\square$

As a consistency check, exact minimal-polynomial computations for $3 \leq d \leq 12$ give precisely the degrees in Theorem 8.2; for example, $C_4 = 9/7$ and $C_6 = (225 + 140\sqrt{70})/961$ [9, Cor. 5.1, Prop. 5.2]. The distinctness statement in Lemma 8.1 was also checked numerically for $4 \leq d \leq 400$.

Corollary 8.3. $[\mathbb{Q}(C_d) : \mathbb{Q}]$ equals $d - 1$ (odd d) or $d/2 - 1$ (even d) for all $3 \leq d \leq 400$, for all even d with $\gcd(v_2(d), d/2 - 1) = 1$ or $d - 1$ prime, and for all $d = 2p + 1$ with p prime and $\text{ord}_p(2) = p - 1$.

9. COMPUTATIONS

All computations use Python 3.12.3, SymPy 1.14.0 and mpmath 1.3.0; the scripts and outputs are in the supplementary archive.

Irreducibility for $3 \leq d \leq 400$. If $A_d = FG$ with $\deg F = k$, then for every prime p not dividing the leading coefficient, k is a sum of a sub-multiset of the degrees of the irreducible factors of $A_d \bmod p$. We intersect these sets over primes until no $k \in [1, m/2]$ survives. The 398 values split as follows: 185 by Theorem 3.1, 6 by Proposition 3.2, 22 by Theorem 4.4, 2 of degree ≤ 1 , and 183 by such certificates (176 odd d and the exceptional $d = 56, 248, 296, 320, 344, 352, 392$). All primes used are ≤ 53 . Exact factorization over $\mathbb{Z}$ for $d \leq 120$ agrees.

Galois groups. For $17 \leq d \leq 82$, $\text{Gal}(A_d) \cong S_m$, certified by Frobenius elements at unramified primes: an m -cycle, a $(m-1, 1)$ -element, an element with a cycle of prime length in $(m/2, m-3]$, and an odd element (Jordan's theorem). For $d = 2p + 1$ and all primes $3 \leq p \leq 199$, $\text{Gal} \cong S_p$: irreducibility holds by the above, Theorem 6.2 applies for $p \geq 5$, and for $p \equiv 7 \pmod{8}$ an odd Frobenius is found at some $q \in \{3, 5, 11\}$.

The family $d = 2p + 2$. For every prime $5 \leq p \leq 3000$ a table records both local sides, with ℓ, a, k , the exact residue $N_k \bmod \ell$, E_ℓ , M_ℓ , and the dual q, b, E_q, M_q . It also records the certificate used and its exact margin. We find 424 of 428 primes certified by local blocks: 381 by the dual block with Jordan–Marggraff, 31 by the ℓ -block with Jordan–Marggraff, and 12 by a dual block with $b \geq 2$ and (d). All 424 are free of CFSG. The minimal margin is $3/14$ for Jordan–Marggraff $(1/2 - s/p)$ and $1/4$ for (d) $(p/4 - s)$. The value $1/4$ occurs exactly when $p + 1 = 2q^b$ (here $p = 53, 337$), since then $s = (p - 1)/4$. The Liebeck–Saxl margin $1/3 - s/p$ (Remark 6.1) is at least $85/1011$. The four exceptions $p \in \{5, 11, 23, 191\}$ are the primes with $p = 3 \cdot 2^s - 1$ and $2p + 1 = 3 \cdot 2^{s+1} - 1$ both prime. For them S_p is certified directly by Frobenius elements.

Parity was settled for every row: Theorem 6.5 in 216 cases, Proposition 6.7 in 161, Proposition 6.8 in 34, and a Stickelberger witness in 13. A Stickelberger witness is a prime q with $(\text{disc } A_d/q) = -1$, which by Stickelberger–Swan [18] gives an odd Frobenius. Here $\text{disc } A_d \bmod q$ is computed as a resultant over $\mathbb{F}_q$, cross-checked against exact discriminants in 366 cases. Hence $\text{Gal}(A_{2p+2}) \cong S_p$ for all primes $3 \leq p \leq 3000$ ($p = 3$: a cubic with non-square discriminant).

For $p \leq 10^6$, Theorem 6.4(1) covers 78482 of 78496 primes. The other 14 are exactly those with $p + 1 \in \{2^s, 3 \cdot 2^s\}$: nine are covered by Theorem 6.4(2), four are the exceptions above, and $p = 524287$ was not checked. The local parity results alone give S_p for 98.2% of these p .

10. OPEN PROBLEMS

- (1) Conjecture 1.1; in particular the *macroscopic* range of factor degrees for odd d . Example 7.3 shows that a global argument is needed for odd prime d .
- (2) The exceptional even set $\mathcal{E}$: find a uniform argument excluding the degrees Q s allowed by Proposition 3.4 (for $g = 3$ only Q and $2Q$). Remark 3.5 shows that local data at the structured primes do so for most, but not all, $d \in \mathcal{E}$.
- (3) Prove $\text{Gal}(A_d) \supseteq A_m$ for composite m . Primitivity is the difficulty; the local cycles of Section 5 restrict block sizes but not the large ones.
- (4) Closed-form square classes for $a \geq 2$, for unclean ℓ , and for q^b with $b \geq 2$.
- (5) Related families: $U_d(2, t) - c$ for integers $c \neq 0$ (irreducible for $c \in \{\pm 1, \pm 2, 3, 4\}$, $d \leq 50$), and the cosine analogue $(V_d(2, t) - 2)/(t - 1)$ with $V_d = \alpha^d + \beta^d$ (the equation $\cos d\theta = \cos^d \theta$; irreducible for $d \leq 50$). In all neighbouring families we tested, reducibility comes only from three mechanisms: a degenerate point $\beta = 0$, α/β a root of unity, or divisibility $U_k \mid U_d$.

SUPPLEMENTARY MATERIAL

Scripts, certificate tables and their checksums are provided in the supplementary archive accompanying this manuscript.

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